

Question	Answer	Marks	Guidance
2(a)(i)	(satellite) remains above the same point on the Earth's surface	B1	Allow 'equator' for 'surface' Ignore references to 'stationary'
2(a)(ii)	<u>Any two points from:</u> • equatorial • west to east • period = 24 hours	B2	List Rules apply. Allow orbit is in same direction as Earth's <u>rotation</u> Allow 'period is same as period of <u>rotation</u> of Earth' Allow '86 400 s' or '1 day' for '24 hours'.
2(b)(i)	$v = 2\pi R / T$	A1	Allow 'r' for 'R'
2(b)(ii)	$GMm / R^2 = mR\omega^2$	M1	
	clear use of $\omega = 2\pi / T$ and completion of algebra to reach $M = (4\pi^2 R^3) / GT^2$	A1	
	or		
	$GMm / R^2 = mv^2 / R$	(M1)	
	clear use of $v = 2\pi R / T$ and completion of algebra to reach $M = (4\pi^2 R^3) / GT^2$	(A1)	

Question	Answer	Marks	Guidance
2(c)	density = mass / volume	C1	<u>All three C1 marks are independent</u> Allow in symbols
	mass = $4\pi^2 \times [(6370 + 35\,900) \times 10^3]^3 / [(6.67 \times 10^{-11}) \times (24 \times 3600)^2]$	C1	$M = 5.988 \times 10^{24}$ kg (to 4 or more SF) implies C1.
	volume = $(4/3)\pi \times (6370 \times 10^3)^3$	C1	$V = 1.083 \times 10^{21}$ m ³ (to 4 or more SF) implies C1.
	average density = 5530 kg m ⁻³	A1	Must be to 3 significant figures. AFC. Allow AFC for 5560 kg m ⁻³ (from $T = 23\text{h}56\text{m}$). An answer given to more than 3 significant figures which rounds to 5530 implies the three C1 marks.